

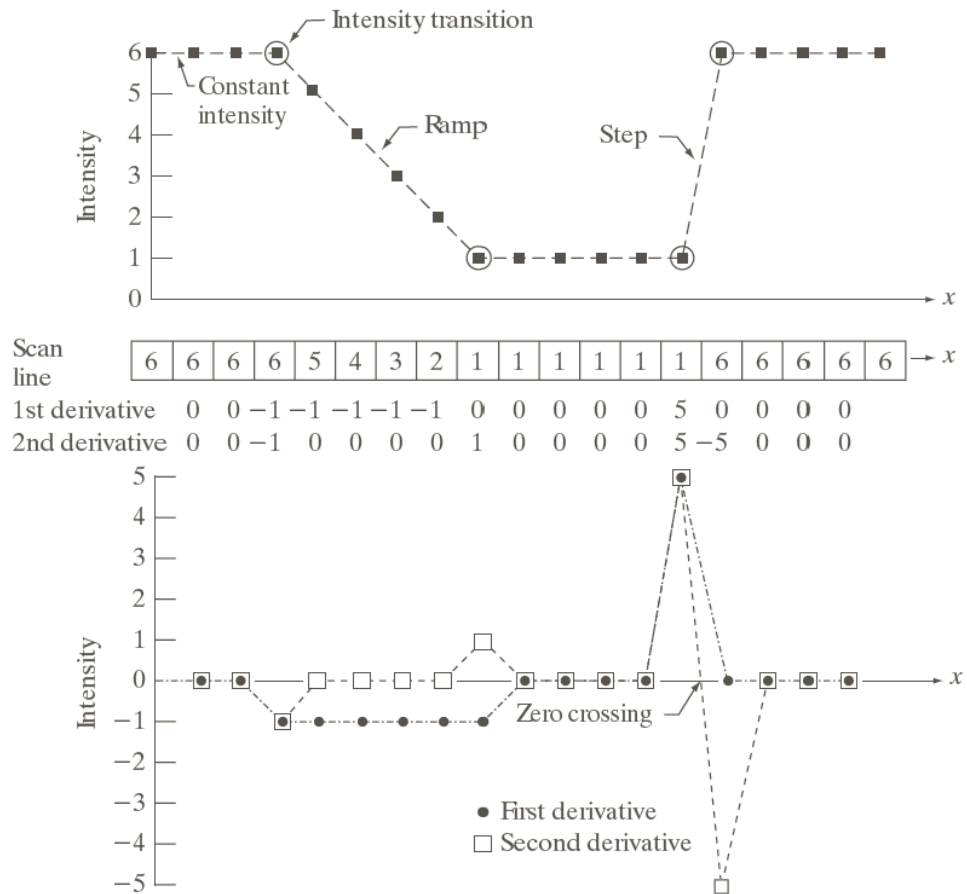
1st and 2nd order derivatives

First derivative
forward difference
(look-ahead)
implementation:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

Second derivative:

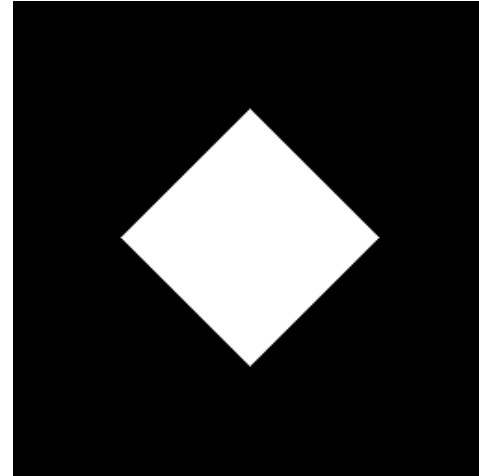
$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \frac{\partial f}{\partial x} = \frac{\partial f}{\partial x}(x) - \frac{\partial f}{\partial x}(x-1) = f(x+1) + f(x-1) - 2f(x)$$



First order derivatives of an image

- The gradient

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$



- Magnitude of the gradient
 - Tells us something about presence of edges

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

- What does the orientation of the gradient vector tell us?

How can we compute first-order discrete image derivatives?

- There are various ways...
 - One dimensional forward differences
 - Roberts cross gradient operators
 - One dimensional central differences
 - Prewitt operators
 - Sobel operators

The *Laplacian*

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

- A single summary measure of second order derivatives
 - Isotropic
- Used for image enhancement as well as edge detection
- Again, various implementations
 - Most implementations are isotropic (rotation invariant) in increments of 45 degrees

0	1	0	1	1	1
1	-4	1	1	-8	1
0	1	0	1	1	1

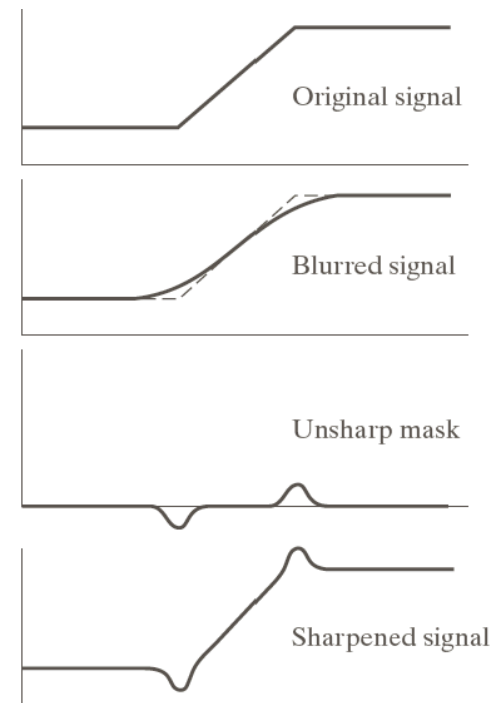
0	-1	0	-1	-1	-1
-1	4	-1	-1	8	-1
0	-1	0	-1	-1	-1

Image enhancement

- Add the Laplacian (scaled by a constant) to the image

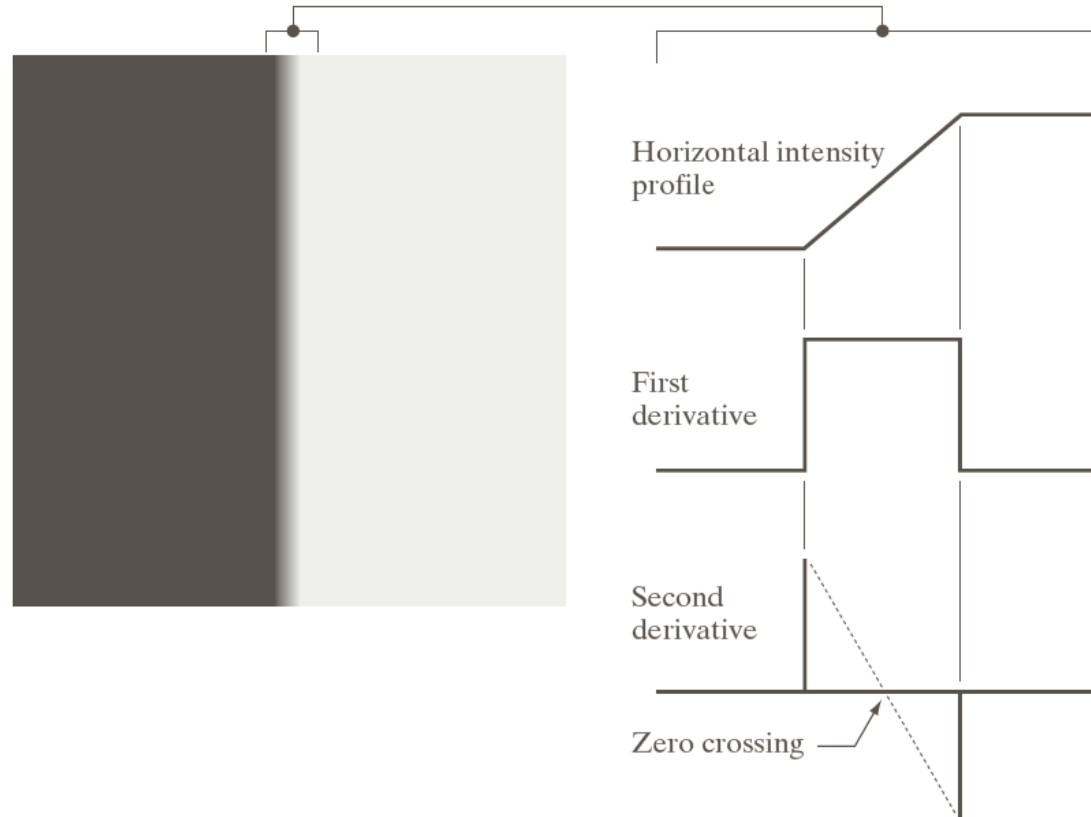
$$g(x,y) = f(x,y) + c\nabla f(x,y)$$

- Unsharp masking
 - Apply averaging filter to f
 - Subtract result from original image
 - Add difference to the original



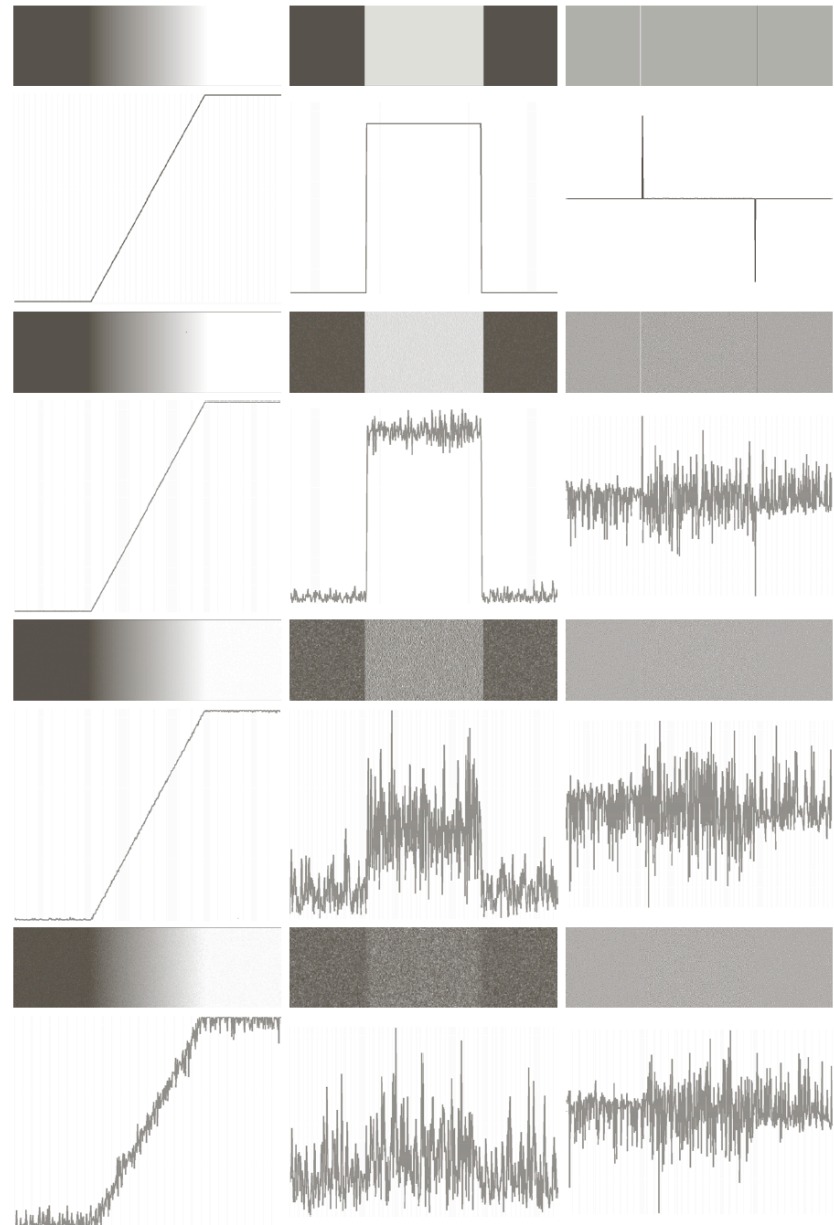
An early preview of edge detection

- Two typical ways to find image edges. Look for:
 - Pixels with large gradient magnitude
 - Zero-crossings of second derivatives

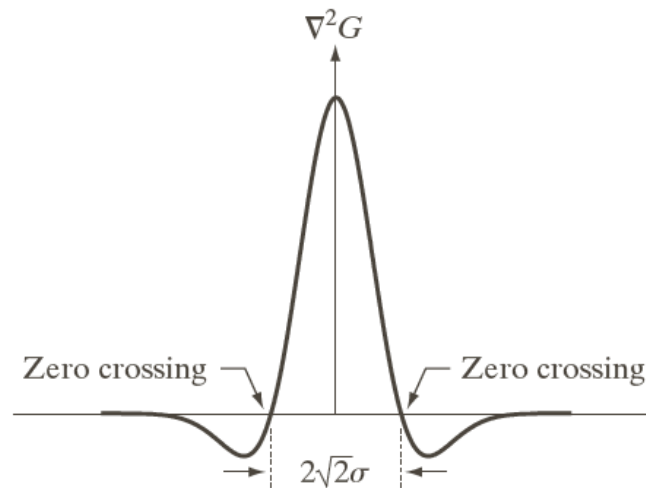
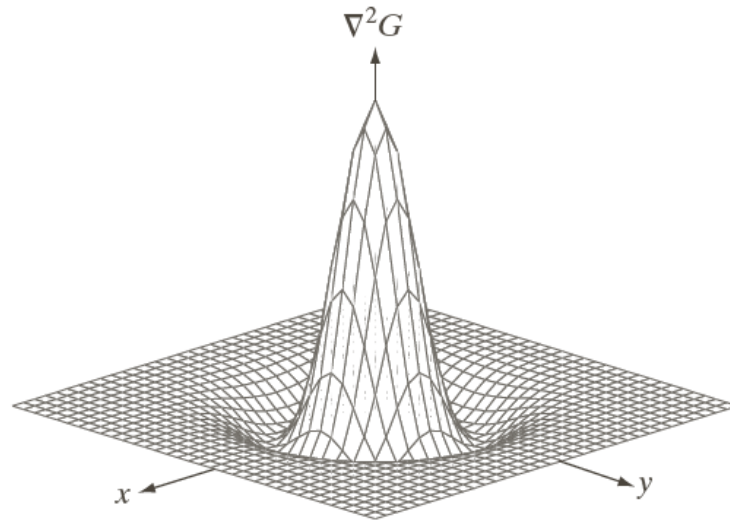


Derivatives in presence of noise

- Differentiation amplifies noise
 - Why?
- How can we get around this?
 - Smooth first
 - Compute derivative later
 - An example is the *Laplacian of Gaussian* (LoG) operator



Laplacian of Gaussian



0	0	-1	0	0
0	-1	-2	-1	0
-1	-2	16	-2	-1
0	-1	-2	-1	0
0	0	-1	0	0

General feature detection

- We can design filter masks to find
 - Edges
 - Lines and curves
 - Corners
 - Other more general structures
- We will come back to this topic later in the semester

Other uses of image derivatives...

- Motion estimation and object tracking in video
- This is another hard problem in computer vision
- We'll take a look at how we can use the image gradient vector and the time derivative to start formulating a solution to these problems
- Specifically, we'll study the basics of the Takeo-Kanade method
 - There are methods such as Horn-Schuck,...

- Read until end of Section 3.7.
- Laplacian of Gaussian is discussed in 10.2.6.
We'll revisit it when we talk about
segmentation and edge detection in more
detail.